Counting half-integral polygons with collinear interior lattice points

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Abstract

We count convex half-integral polygons with a prescribed number of collinear interior lattice points, up to affine unimodular equivalence. For every $i \geq 3$, the count is the degree-seven polynomial conjectured by Bohnert and Springer. The proof is computer-assisted: their strip theorem reduces the polygons to two boundary chains on five integer rows, and a unique integer shear normalization gives finitely many integer linear systems. Complete rational generating functions for these systems, together with the fixed classes of the remaining reflections, give the formula by Burnside's lemma. Integral polygons are included, and translations are integral.

1 Introduction

A half-integral polygon is a two-dimensional convex polygon in $\mathbb{R}^2$ whose vertices lie in $\frac{1}{2}\mathbb{Z}^2$. This includes lattice polygons. We identify P and P' when

$$P' = UP + t, \qquad U \in \mathrm{GL}_2(\mathbb{Z}), \quad t \in \mathbb{Z}^2.$$

For an integer $i \geq 3$, let $N(i)$ denote the number of these equivalence classes having exactly i interior points of $\mathbb{Z}^2$, all on one line. Prescribing the denominator and the interior lattice points gives a concrete classification problem for rational polygons. In the collinear case, the line of interior points also supplies coordinates suited to counting.

Bohnert and Springer [1] classify rational polygons with small denominator and few interior lattice points. Their Proposition 3.5 puts a k-rational polygon with collinear interior lattice points in $\mathbb{R} \times [-1, 1]$ when the number of those points exceeds k. Here k-rational means that multiplication by k makes every vertex integral; the denominator need not be exactly k. Their Classification 3.6 records the half-integral counts for $3 \leq i \leq 20$, and their Conjecture 3.7 gives the polynomial below. We give a computer-assisted proof over the whole stated range.

Theorem 1.1. *For every integer $i \geq 3$,*

$$\begin{aligned} N(i) = \frac{i+1}{1260}\big(&512i^6 + 12928i^5 + 137740i^4 + 685145i^3 \\ &+ 1582743i^2 + 1665222i + 710640\big). \end{aligned} \tag{1.1}$$

Here $N(i)$ counts two-dimensional convex polygons with vertices in $\frac{1}{2}\mathbb{Z}^2$, including integral polygons, with exactly i collinear interior points of $\mathbb{Z}^2$, modulo $\mathrm{GL}_2(\mathbb{Z})$ and translations in $\mathbb{Z}^2$.

The strip theorem leaves only five possible vertex rows after doubling. It does not by itself remove the infinitely many integer shears. Normalizing one boundary slope removes those shears

uniquely and leaves finitely many coordinate choices at each i. The remaining equivalences form a four-element group. Its stabilizers matter, so counting normalized polygons and simply dividing by four would be incorrect. We instead compute the generating functions of the four fixed sets. Each is a finite sum of integer-point generating functions, with the number of interior points retained as an unbounded variable.

The rational-series calculation uses Normaliz through PyNormaliz [2, 3, 4]. The proof therefore relies on trusted exact software for a finite collection of complete generating functions. The geometric correspondence, the fixed-point equations and the final coefficient identity are given below. The accompanying computational supplement contains the exact systems, complete rational outputs and reproduction commands.

Section 2 gives the boundary description and shear normalization. Section 3 determines the remaining equivalences and fixed sets. Section 4 performs the generating-function count and recovers (1.1). Throughout, lattice points mean points of $\mathbb{Z}^2$; only integer translations of the original polygon are allowed.

2 Normalized boundary chains

2.1 The interior row and the doubled polygon

We use the following published result with its original parameter range.

Proposition 2.1 (Bohnert–Springer, Proposition 3.5)**.** *A two-dimensional convex k-rational polygon with collinear interior lattice points and more than k such points is affinely unimodularly equivalent to a polygon contained in $\mathbb{R} \times [-1, 1]$.*

For its proof see [1, pp. 1248–1249]. Apply Proposition 2.1 with $k = 2$. The only integer horizontal row strictly inside the strip is $y = 0$. The interior lattice points on that row are consecutive: convexity puts every point between the first and last in the interior. After an integer horizontal translation, they are

$$(j, 0), \qquad 1 \le j \le i.$$

Set $n = i + 1 \ge 4$ and $Q = 2P$. Then Q has integral vertices and its interior points of the even lattice $2\mathbb{Z}^2$ are exactly

$$(2j, 0), \qquad 1 \le j \le n - 1. \tag{2.1}$$

The lowest and highest vertex heights of Q are $-a$ and b, with $a, b \in \{1, 2\}$, since the interior row lies strictly between them.

Write the horizontal sections as

$$Q \cap (\mathbb{R} \times \{y\}) = [f(y),\, 2n - g(y)] \times \{y\}, \qquad -a \le y \le b. \tag{2.2}$$

The left boundary coordinate f and the right boundary distance g are convex piecewise affine functions. Their breakpoints have integer heights and integer values. Each boundary's height set contains $-a, b$ and any subset of the intermediate integer heights. A height is included in this set precisely when the boundary has a strict change of slope there. The left and right height sets can differ. At an extreme height the horizontal face may have length zero.

2.2 Exact integer conditions

The row condition (2.1) is equivalent to

$$0 \le f(0) < 2, \qquad 0 \le g(0) < 2. \tag{2.3}$$

Indeed, the interior row section must contain $2, 4, \dots, 2n-2$ strictly and must not contain 0 or $2n$ strictly. For three successive boundary heights $u < v < w$, write the boundary values as x_u, x_v, x_w. Strict slope increase is equivalent to the integer inequality

$$(w-v)x_u - (w-u)x_v + (v-u)x_w \ge 1. \tag{2.4}$$

For the unique consecutive heights $u \le 0 < v$ containing the right side of zero, put $d = v - u$. Linear interpolation gives

$$x(0) = \frac{vx_u - ux_v}{d}.$$

Thus the corresponding intercept condition is

$$0 \le vx_u - ux_v \le 2d - 1. \tag{2.5}$$

This applies separately to each boundary, with its own central edge. The boundaries do not cross if and only if

$$f(-a) + g(-a) \le 2n, \qquad f(b) + g(b) \le 2n. \tag{2.6}$$

To see sufficiency, $f + g$ is convex, hence is no larger than the larger of its endpoint values.

Lemma 2.2. *For fixed $n \ge 4$, conditions (2.4), (2.5) and (2.6), with integer boundary values and the stated height sets, describe exactly the normalized polygons Q satisfying (2.1). The description is unique.*

Proof. For admitted chain data let

$$K = \{(x, y) : -a \le y \le b,\ f(y) \le x \le 2n - g(y)\}.$$

This is convex, being the intersection of the epigraph of f with the hypograph of $2n - g$ and the height strip. Its width $w(y) = 2n - f(y) - g(y)$ is nonnegative by (2.6) and is concave. At zero, $w(0) > 2n - 4 \ge 4$. Interpolating between zero and either endpoint shows that w is positive at every nonendpoint height. Hence K is two-dimensional. Its vertices are the strict side breakpoints and the endpoints of the extreme faces, all integral. A point extreme face contributes one vertex, although the point belongs to both chains.

Condition (2.3) gives exactly the interior even-lattice points in (2.1). The only other possible even rows are $y = -2, 2$, and each is absent or is an extreme boundary row. Thus $K/2$ is an admissible half-integral polygon. Conversely, every normalized polygon has these boundaries and satisfies the conditions. Strict slope increase excludes redundant collinear vertices, so its height sets and coordinates are recoverable uniquely. This also retains corners at height zero and zero-length extreme faces. □

2.3 Removing integer shears

An integer shear is

$$S_s(x, y) = (x + sy, y), \qquad s \in \mathbb{Z}.$$

It sends (f, g) to $(f + sy, g - sy)$. We choose the unique representative with

$$0 \le f'_+(0) < 1. \tag{2.7}$$

Adding an integer to the right derivative puts it in this half-open interval in exactly one way. If (u, v) is the left central edge, condition (2.7) says

$$0 \le x_v - x_u \le d - 1, \qquad d = v - u.$$

For $\delta = x_v - x_u$ the reference values satisfy

$$\delta \in \{0, \dots, d-1\}, \qquad 0 \le dx_u - u\delta < 2d, \qquad x_v = x_u + \delta. \tag{2.8}$$

For each δ, the half-open interval for x_u has length two and contains exactly two integers. There are therefore $2d$ reference pairs. Also $x_u = f(0) + (u/d)\delta \in (-2, 2)$, so the pairs can be found by checking $x_u = -2, -1, 0, 1, 2$ and retaining exactly (2.8).

Lemma 2.3. *For every $n \ge 4$ there are finitely many normalized chain data satisfying (2.7). The integer shear action on the polygons is free.*

Proof. The right derivative of a convex piecewise affine function is a supporting slope at zero, even when zero is a corner. Consequently

$$f(y) \ge f(0) + f'_+(0)y \ge -2.$$

The endpoint widths give $g(-a), g(b) \le 2n + 2$. Convexity gives the same upper bound throughout. For $y < 0$, interpolation of g between y and b yields

$$g(y) \ge \frac{b - y}{b} g(0) + \frac{y}{b} g(b) \ge -4n - 4.$$

For $y > 0$, interpolation between $-a$ and y gives

$$g(y) \ge \frac{a + y}{a} g(0) - \frac{y}{a} g(-a) \ge -4n - 4.$$

The endpoint widths now bound the endpoint values of f above by $6n + 4$, and convexity gives this upper bound throughout. All boundary coordinates thus lie in bounded intervals depending linearly on n. There are finitely many height sets, so the integer fiber at n is finite. The same estimates bound the real fibers.

If $S_s Q = Q$ with $s \ne 0$, choose a point of Q at nonzero height. Its iterated shear images have unbounded horizontal coordinates, contrary to boundedness of Q. Hence no nonzero shear fixes Q. □

3 Equivalences and fixed classes

3.1 The residual four-element group

An equivalence between two normalized original polygons must preserve the set $\{(j, 0) : 1 \le j \le n-1\}$. This set determines its line, so the linear part of the equivalence has the form

$$\begin{pmatrix} \varepsilon & s \\ 0 & \eta \end{pmatrix}, \qquad \varepsilon, \eta \in \{1, -1\}, \quad s \in \mathbb{Z}.$$

Preserving the first and last interior points forces the translation to be $(0,0)$ if $\varepsilon = 1$ and $(n,0)$ if $\varepsilon = -1$. In doubled coordinates the latter translation is $(2n,0)$. Thus all normalized equivalences are generated by the shears and

$$h(x,y) = (2n - x, y), \qquad v(x,y) = (x,-y). \tag{3.1}$$

Both h and v conjugate S_s to S_{-s}, and hv commutes with all shears. Quotienting by the normal shear subgroup leaves the Klein four group $G = \{e, h, v, hv\}$.

Let X_n be the finite set of shear classes at n, represented by (2.7). The original polygon classes are exactly the G-orbits in X_n. Every action preserves n, including the affine map h.

3.2 Fixed shear classes

If a shear class is fixed by $\sigma \in G$, then $\sigma Q = S_m Q$ for an integer m. By freeness in Lemma 2.3, this integer is unique. Comparing the two boundary functions gives the following conditions.

For h, the left and right height sets must agree and

$$g(y) = f(y) + my. \tag{3.2}$$

Indeed, hQ has left coordinate $g(y)$, whereas $S_m Q$ has left coordinate $f(y)+my$; the right-coordinate equality is the same equation.

For v, one must have $a = b$, each height set invariant under negation, and

$$f(-y) = f(y) + my, \qquad g(-y) = g(y) - my. \tag{3.3}$$

The equality $vQ = S_m Q$ compares sections at y and $-y$. These equations need only be imposed at the paired vertex heights: both sides are affine on the same intervening intervals.

For hv, one must have $a = b$ and the right height set equal to the negative of the left one. If $hvQ = S_m Q$, applying hv twice and using commutation with shears gives $Q = S_{2m} Q$. Freeness forces $m = 0$. The remaining condition is

$$g(-y) = f(y). \tag{3.4}$$

Conversely, these respective conditions give $hQ = S_m Q$, $vQ = S_m Q$, or $hvQ = Q$, and hence a fixed shear class. We impose them on the representatives with the unique left slope normalization; they do not require the polygon itself to be fixed before taking the shear quotient.

For each $\sigma \in G$ define the formal generating function

$$F_\sigma(z) = \sum_{n \geq 4} |\operatorname{Fix}_{X_n}(\sigma)| z^n.$$

These coefficients are finite by Lemma 2.3. Burnside's formula gives

$$C(z) := \sum_{n \geq 4} N(n-1) z^n = \frac{F_e(z) + F_h(z) + F_v(z) + F_{hv}(z)}{4}. \tag{3.5}$$

For completeness, count pairs (σ, x) with $\sigma x = x$. An orbit with stabilizer size s has $4/s$ elements and contributes $(4/s)s = 4$ pairs. Dividing the total by four counts the orbits, including those with nontrivial stabilizers.

4 Exact generating functions and the count

4.1 The finite systems

The boundary description gives explicit integer linear systems with n as an unbounded integer variable. For e, enumerate $(a, b) \in \{1, 2\}^2$, both height subsets containing the endpoints, and every left reference pair in (2.8). Fix those two left values; keep the other side coordinates and n as variables. Impose $n \geq 4$, all strict slope inequalities (2.4), both intercept inequalities (2.5), and the endpoint widths (2.6).

For h, require equal height sets and substitute (3.2), keeping its integer m as a variable. For v, require $a = b$ and the two symmetric height sets, and add (3.3) as integer equations. For hv, require $a = b$, reflect the left height set for the right one, and substitute (3.4). All cases retain the same left reference-pair rule and all boundary conditions. The integer m, when present, is unique and bounded at each n by the boundary coordinates at a nonzero height.

These loops give the disjoint cells in Table 1. Disjointness follows from the unique height sets, reference pair and shear representative. In a fixed set it also uses uniqueness of m. Every integer point of a cell gives one of the appropriate fixed classes, and every fixed class belongs to exactly one cell.

(a, b)	e	h	v	hv
$(1, 1)$	12	6	12	6
$(1, 2)$	64	16	0	0
$(2, 1)$	56	14	0	0
$(2, 2)$	288	36	72	36
Total	420	72	84	42

Table 1: Number of integer linear systems in each fixed-class count. These are input cells, not numbers of polygons. Empty cells contribute zero.

The supplement's `src/chain_models.py` implements precisely these loops. A row $[a_1, \ldots, a_r, c]$ encodes $\sum_j a_j x_j + c \geq 0$ or $= 0$, with every $x_j \in \mathbb{Z}$. The grading is the unit vector selecting the original n coordinate. No coordinate-sign or parity restriction is added. There are 618 cell members but 608 literally distinct complete matrix and grading inputs. Equal complete inputs share a computed rational function; every original cell retains its multiplicity in the sum.

4.2 The all-parameter series calculation

For a cell with inequalities $Ax + c \geq 0$ and equations $Bx + d = 0$, homogenization replaces the constants by $c\lambda, d\lambda$ and adds $\lambda \geq 0$. Its integer points at $\lambda = 1$ are exactly the original integer solutions. The n grading extends by zero on the homogenizing coordinate. The real degree fibers are bounded by the estimates in Lemma 2.3; the fixed-class equations do not change this finiteness. Thus the inhomogeneous integer-point series is well defined.

Normaliz computes these rational series by exact rational-cone enumeration [2, Section 5.3]. A disjoint semi-open simplicial decomposition and the integer points in each fundamental domain give monomials divided by the products of $1 - z^{d_j}$ belonging to the generating rays. Extracting the $\lambda = 1$ slice gives the inhomogeneous series. The integer-lattice, row and grading conventions used here are described in [3, Sections 2.1, 2.9, 4.1.18–4.1.19, 4.5.1 and 4.12–4.13]. In particular, the explicit inequalities prevent the default positive orthant convention from restricting signed coordinates, and

equations are intersected with the full integer lattice. Cells in proper affine subspaces and empty cells are included.

The calculation uses PyNormaliz 2.24 and the collective request

`Compute(HilbertSeries, BigInt, NoGradingDenom).`

Each returned grading is checked to be the original n unit vector, with homogenizing zero and denominator one. `NoGradingDenom` preserves this grading, including affine lattices with nontrivial period. The result is a complete rational function over all $n \geq 4$; no coefficient cutoff is used in its computation.

Lemma 4.1 (Computer-assisted rational-series calculation)**.** *Summing the complete integer-point series of the cells in Table 1, with all original multiplicities, gives*

$$F_e(z) = \frac{z^4}{(1-z)^8}(669807 - 3348976z + 7783173z^2 - 10577048z^3 + 8928985z^4 - 4635736z^5 + 1361923z^6 - 173936z^7), \tag{4.1}$$

$$F_h(z) = \frac{z^4(2349 - 7519z + 9303z^2 - 5255z^3 + 1138z^4)}{(1-z)^5}, \tag{4.2}$$

$$F_v(z) = \frac{z^4(2069 - 6465z + 7915z^2 - 4447z^3 + 960z^4)}{(1-z)^5}, \tag{4.3}$$

$$F_{hv}(z) = \frac{z^4(335 - 806z + 669z^2 - 190z^3)}{(1-z)^4}. \tag{4.4}$$

Computational verification. The supplement records the exact input, whole rational output and original grading for each of the 608 distinct systems, and the assignment of all 618 cell members to those systems. All systems were computed in ordinary `BigInt` mode. An integer-only driver sums the functions with their cell multiplicities: the returned denominator degrees are one or two, so $1 - z^2 = (1 - z)(1 + z)$ gives a common denominator using integer polynomial multiplication and addition. The four rational differences from (4.1)–(4.4) are zero. The driver checks full numerator identities rather than a finite number of power-series coefficients. □

This lemma is the proof's software-dependent step. The supplement pins the actual wheel and native-library bytes. According to [3, Section 5.3 and Appendix E.2.10], `BigInt` avoids the initial 64-bit main computation. It does not enlarge bounded degree types, and ordinary LLL arithmetic remains part of the implementation. The computation completed normally; its observed output degrees do not bound every intermediate operation. The manual version is 3.11.1, while the exact minor version of the compiled Normaliz library in the wheel is not exposed and is not inferred from it. The calculation relies on this trusted software and runtime; an independent counting implementation or a separately checked simplicial-decomposition certificate is not supplied.

4.3 Recovering all coefficients

Combining Lemma 4.1 with (3.5) gives

$$C(z) = \frac{z^4 P_*(z)}{(1-z)^8}, \tag{4.5}$$

where the numerator polynomial is

$$P_*(z) = 168640 - 844590z + 1965375z^2 - 2673454z^3 + 2258540z^4 - 1173244z^5 + 344837z^6 - 44056z^7. \tag{4.6}$$

This is an identity of rational functions, hence of their formal expansions at zero.

Proof of Theorem 1.1. Write c_k for the coefficient of z^k in (4.6). The formal geometric-series identity gives

$$(1-z)^{-8} = \sum_{r\geq 0} \binom{r+7}{7} z^r.$$

Since the degree variable is $n = i+1$, extracting z^{i+1} in (4.5) yields

$$N(i) = \sum_{k=0}^{7} c_k \binom{i+4-k}{7}, \qquad i \geq 3. \tag{4.7}$$

If a shifted exponent is negative, its coefficient is zero and the corresponding binomial upper argument is in $\{0,\dots,6\}$. The polynomial product defining that binomial also vanishes. The smallest upper argument is $i-3 \geq 0$, so this formula includes every endpoint case in the stated range.

Multiplying (4.7) by $7! = 5040$ and expanding gives the integer polynomial identity

$$\begin{aligned}\sum_{k=0}^{7} c_k \prod_{j=0}^{6} (i+4-k-j) = 4(i+1)\big(&512i^6 + 12928i^5 + 137740i^4 + 685145i^3 \\ &+ 1582743i^2 + 1665222i + 710640\big).\end{aligned} \tag{4.8}$$

Both sides have ascending coefficients

$$(2842560,\ 9503448,\ 12991860,\ 9071552,\ 3291540,\ 602672,\ 53760,\ 2048).$$

Dividing by 5040 proves (1.1), since $5040 = 4 \cdot 1260$. The degree-seven identity follows from the complete rational series; finite classification data are not used to infer it. □

Computational materials. The accompanying versioned supplement contains the chain generator, the complete compact matrix/series data, an integer-only aggregation verifier, a Normaliz replay driver, finite integer-geometry controls and environment information. Its README distinguishes verification of the stored algebra from recomputation of the software-dependent series. The full replay is optional for using the supplied data and is required to reproduce that series calculation from the inputs.

AI-use statement. OpenAI Codex was used to develop the boundary encoding, implement and run the exact computations, check the argument and prepare the manuscript. Any accompanying manuscript review identifies its AI reviewer and the reviewed version. AI review and successful software tests do not constitute independent human mathematical certification.

References

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